1. (10 points) **Noisy linear regression**

A real estate company have assigned us the task of building a model to predict the house prices in Westwood. For this task, the company has provided us with a dataset $\mathcal{D}$:

$$\mathcal{D} = \{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(N)}, y^{(N)})\}$$

where $x^{(i)} \in \mathbb{R}^d$ is a feature vector of the ith house and $y^{(i)} \in \mathbb{R}$ is the price of the ith house. Since we just learned about linear regression, we have decided to use a linear regression model for this task. Additionally, the IT manager of the real estate company have requested us to design a model with small weights. In order to accommodate his request, we will design a linear regression model with parameter regularization. In this problem, we will navigate through the process of achieving regularization by adding noise to the feature vectors. Recall, that we define the cost function in a linear regression problem as:

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(y^{(i)} - (x^{(i)})^T\theta)^2$$

where $\theta \in \mathbb{R}^d$ is the parameter vector. As mentioned earlier, we will be adding noise to the feature vectors in the dataset. Specifically, we will be adding zero-mean gaussian noise of known variance σ^2 from the distribution

$$\mathcal{N}(0, \sigma^2 I)$$

where $I \in \mathbb{R}^{d\times d}$ and $\sigma \in \mathbb{R}$. With the addition of gaussian noise the modified cost function is given by

$$\tilde{\mathcal{L}}(\theta) = \frac{1}{N}\sum_{i=1}^{N}(y^{(i)} - (x^{(i)} + \delta^{(i)})^T\theta)^2$$

μ : Mean Vector

where $\delta^{(i)}$ are i.i.d noise vectors with $\delta^{(i)} \sim \mathcal{N}(0, \sigma^2 I)$. Covariance Matrix

Independent and identically distributed

(a) (6 points) Express the expectation of the modified loss over the gaussian noise, $\mathbb{E}_{\delta\sim\mathcal{N}}[\tilde{\mathcal{L}}(\theta)]$, in terms of the original loss plus a term independent of the data $\mathcal{D}$. To be precise, your answer should be of the form:

$$\mathbb{E}_{\delta\sim\mathcal{N}}[\tilde{\mathcal{L}}(\theta)] = \mathcal{L}(\theta) + R$$

where R is not a function of $\mathcal{D}$. For answering this part, you might find the following result useful:

$$\mathbb{E}_{\delta\sim\mathcal{N}}[\delta\delta^T] = \sigma^2 I$$

(a)

$$(y^{(i)} - (x^{(i)} + \delta^{(i)})^T\theta)^2 = (y^{(i)} - (x^{(i)})^T\theta - (\delta^{(i)})^T\theta)^2$$

$$= \underbrace{(y^{(i)} - (x^{(i)})^T\theta)^2}_{A} + \underbrace{((\delta^{(i)})^T\theta)^2}_{B} - \underbrace{2(y^{(i)} - (x^{(i)})^T\theta)(\delta^{(i)})^T\theta}_{C}$$

By linearity of expectation : $E[\sum_{i=1}^{N} x_i] = \sum_{i=1}^{N} E[x_i]$, and $E[\cdot]$ is a linear operation

$$\therefore E_{\delta\sim N}[\tilde{L}(\theta)] = \frac{1}{N}\sum_{i=1}^{N}\left(E_{\delta\sim N}[A] + E_{\delta\sim N}[B] - E_{\delta\sim N}[C]\right)$$

① Since A has no δ dependence,

$$\therefore E_{\delta\sim N}[A] = (y^{(i)} - (x^{(i)})^T\theta)^2$$

②

$$E_{\delta\sim N}[(\delta^{(i)T}\theta)^2] = E_{\delta\sim N}[(\delta^{(i)T}\theta)^T(\delta^{(i)T}\theta)]$$

$$= E_{\delta\sim N}[\theta^T\delta^{(i)}\delta^{(i)T}\theta]$$

$$= \theta^T \underbrace{E[\delta^{(i)}\delta^{(i)T}]}_{\sigma^2 I}\theta$$

$$= \sigma^2\|\theta\|_2^2$$

$\delta^{(i)} \sim N(0, \sigma^2 I) \rightarrow E[\delta] = 0$

$\therefore Cov[\delta^{(i)}\delta^{(i)T}] = E[\delta^{(i)}\delta^{(i)T}] - \cancel{E[\delta^{(i)}]}^{0}\cancel{E[\delta^{(i)T}]}^{0}$

③

$$E_{\delta\sim N}[2(y^{(i)} - x^{(i)T}\theta)(\delta^{(i)T}\theta)] = 2(y^{(i)} - x^{(i)T}\theta)\underbrace{E_{\delta\sim N}[\delta^{(i)T}\theta]}_{\rightarrow\ \theta^T\delta^{(i)}}$$

(for a scalar k, $k = k^T$)

$$= 2(y^{(i)} - x^{(i)}\theta)\theta^T \cancel{E_{\delta\sim N}[\delta^{(i)}]}^{0} = 0$$

$$\therefore E_{\delta\sim N}[\tilde{L}(\theta)] = \frac{1}{N}\sum_{i=1}^{N}\left[(y^{(i)} - x^{(i)T}\theta)^2 + \sigma^2\|\theta\|_2^2\right]$$

$$= \underbrace{\frac{1}{N}\sum_{i=1}^{N}(y^{(i)} - x^{(i)T}\theta)^2}_{L(\theta)} + \underbrace{\frac{1}{N}\cdot N\cdot\sigma^2\|\theta\|_2^2}_{R} \longrightarrow R = \sigma^2\|\theta\|_2^2 \quad \#$$

$E[\tilde{L}(\theta)]$

(b) (2 points) Based on your answer to (a), under expectation what regularization effect would the addition of the noise have on the model?

same as disscussion note.

(c) (1 point) Suppose $\sigma \longrightarrow 0$, what effect would this have on the model?

(d) (1 point) Suppose $\sigma \longrightarrow \infty$, what effect would this have on the model?

(b) L2 regularization (Ridge regression) : $\sigma^2\|\theta\|_2^2$ penalizes large weights to prevent overfitting.

(c) As the regularization strength $\sigma \to 0$, we will have no regulization.
This might let the model "overfit" to the dataset.

(d) As the regularization strength $\sigma \to \infty$, the loss function objective is dominated by $\sigma^2\|\theta\|_2^2$ term.
This will now focus on minimizing the L2 norm of the parameters vector θ.
This will make $\theta \to 0$ and the model doesn't learn much from the data ("underfitting")

3. (30 points) **Softmax classifier gradient.** For softmax classifier, derive the gradient of the log likelihood.

Concretely, assume a classification problem with c classes

- Samples are $(\mathbf{x}^{(1)}, y^{(1)}), \ldots, (\mathbf{x}^{(m)}, y^{(m)})$, where $\mathbf{x}^{(j)} \in \mathbb{R}^n$, $y^{(j)} \in \{1, \ldots, c\}$, $j = 1, \ldots, m$
- Parameters are $\theta = \{\mathbf{w}_i, b_i\}_{i=1,\ldots,c}$
- Probablistic model is

$$\Pr\left(y^{(j)} \mid \mathbf{x}^{(j)}, \theta\right) = \text{softmax}_{y^{(j)}}(\mathbf{x}^{(j)})$$

where

$$\text{softmax}_{y^{(j)}}(\mathbf{x}^{(j)}) = \frac{e^{\mathbf{w}_{y^{(j)}}^T \mathbf{x}^{(j)} + b_{y^{(j)}}}}{\sum_{k=1}^{c} e^{\mathbf{w}_k^T \mathbf{x}^{(j)} + b_k}}$$

Derive the log-likelihood $\mathcal{L}$, and its gradient w.r.t. the parameters, $\nabla_{\mathbf{w}_i}\mathcal{L}$ and $\nabla_{b_i}\mathcal{L}$, for $i = 1, \ldots, c$.

Hint1: We can group $\mathbf{w}_i$ and b_i into a single vector by augmenting the data vectors with an additional dimension of constant 1. Let $\tilde{\mathbf{x}} = \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}$, $\tilde{\mathbf{w}}_i = \begin{bmatrix} \mathbf{w}_i \\ b_i \end{bmatrix}$, then $a_i(\mathbf{x}) = \mathbf{w}_i^T\mathbf{x} + b_i = \tilde{\mathbf{w}}_i^T\tilde{\mathbf{x}}$. This unifies $\nabla_{\mathbf{w}_i}\mathcal{L}$ and $\nabla_{b_i}\mathcal{L}$ into $\nabla_{\tilde{\mathbf{w}}_i}\mathcal{L}$. You can use $\nabla_{\tilde{\mathbf{w}}_i}\mathcal{L}$ as the final answer.

Hint2: You only need to derive the expression for $\nabla_{\tilde{\mathbf{w}}_i}\mathcal{L}$ once for a general index i. This expression is identical for all classes $i \in \{1, \ldots, c\}$.

Hint3: An Indicator function takes on the value of 1 at certain designated points and 0 at all other points. Mathematically, we can represent it as follows:

$$\mathbb{1}_{\{i=y^{(j)}\}} = \begin{cases} 1, & \text{if } i = y^{(j)} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

This can be handy in the final expression for $\nabla_{\tilde{\mathbf{w}}_i}\mathcal{L}$.

Let $\tilde{x}^{(j)} = \begin{bmatrix} x^{(j)} \\ 1 \end{bmatrix}$, $\tilde{w}_i = \begin{bmatrix} w_i \\ b_i \end{bmatrix} \rightarrow w_i^T x^{(j)} + b_i = \tilde{w}_i^T \tilde{x}^{(j)}$

The Log-Likelihood L for m samples is

$$L = \sum_{j=1}^{m} \log \Pr(y^{(j)} | x^{(j)}, \theta) = \sum_{j=1}^{m} \log\left(\frac{e^{\tilde{w}_{y^{(j)}}^T \tilde{x}^{(j)}}}{\sum_{k=1}^{c} e^{\tilde{w}_k^T \tilde{x}^{(j)}}}\right) = \sum_{j=1}^{m}\left(\underbrace{\tilde{w}_{y^{(i)}}^T \tilde{x}^{(j)}}_{A} - \underbrace{\log \sum_{k=1}^{c} e^{\tilde{w}_k^T \tilde{x}^{(j)}}}_{B}\right)$$

A.

$$\frac{\partial}{\partial \tilde{w}_i}\left(\tilde{w}_{y^{(j)}}^T \tilde{x}^{(j)}\right) = \mathbb{1}_{\{i = y^{(j)}\}} \tilde{x}^{(j)}$$

B.

$$\frac{\partial}{\partial \tilde{w}_i}\left(\log \sum_{k=1}^{c} e^{\tilde{w}_k^T \tilde{x}^{(j)}}\right) = \frac{1}{\sum_{k=1}^{c} e^{\tilde{w}_k^T \tilde{x}^{(j)}}} \cdot \underbrace{\frac{\partial}{\partial \tilde{w}_i}\left(\sum_{k=1}^{c} e^{\tilde{w}_k^T \tilde{x}^{(j)}}\right)}_{\text{only } e^{\tilde{w}_i^T x^{(j)}} \text{ left}} = \frac{e^{\tilde{w}_i^T \tilde{x}^{(j)}}}{\sum_{k=1}^{c} e^{\tilde{w}_k^T \tilde{x}^{(j)}}} \tilde{x}^{(j)} = \Pr(y^{(j)} = i \mid x^{(j)}, \theta)\, \tilde{x}^{(j)}$$

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

$$\therefore \nabla_{\tilde{w}_i} L = \sum_{j=1}^{m}\left(\mathbb{1}_{\{i = y^{(j)}\}} - \Pr(y^{(j)} = i \mid x^{(j)}, \theta)\right) \tilde{x}^{(j)} \quad \#$$

4. (10 points) **Hinge loss gradient.**

Due to the drastic changes in climate throughout the world, a weather forecasting organization wants our help to build a model that can classify the observed weather patterns as severe or not severe. They have accumulated data on various attributes of the weather pattern such as temperature, precipitation, humidity, wind speed, air pressure, and geographical location along with severity of weather. However, the contribution of the attributes to the classification of weather as severe or not is unknown.

We choose to use a binary support vector machine (SVM) classification model. The SVM model parameters are learned by optimizing a hinge loss. The company has provided us with a dataset

$$\mathcal{D} = \{(\mathbf{x}^{(1)}, y^{(1)}), (\mathbf{x}^{(2)}, y^{(2)}), \cdots, (\mathbf{x}^{(K)}, y^{(K)})\}$$

where $\mathbf{x}^{(i)} \in \mathbb{R}^d$ is a feature vector of the ith data sample and $y^{(i)} \in \{-1, 1\}$. We define the hinge loss per training sample as

≥ 1 ☺ (loss = 0)

$$\text{hinge}_{y^{(i)}}(\mathbf{x}^{(i)}) = \max\left(0, 1 - y^{(i)}(\mathbf{w}^T\mathbf{x}^{(i)} + b)\right) \tag{2}$$

< 1 ☹

, where $\mathbf{w} \in \mathbb{R}^d$ and bias $b \in \mathbb{R}$ are the model parameters. With the hinge loss per sample defined, we can then formulate the average loss for our model as:

$$\mathcal{L}(\mathbf{w}, b) = \frac{1}{K}\sum_{i=1}^{K} \text{hinge}_{y^{(i)}}(\mathbf{x}^{(i)}) \tag{3}$$

Find the gradient of the loss function $\mathcal{L}(\mathbf{w}, b)$ with respect to the parameters i.e $\nabla_{\mathbf{w}}\mathcal{L}$ and $\nabla_b\mathcal{L}$.

Hint: The Indicator function is also useful here.

$$L(w,b) = \frac{1}{K}\sum_{i=1}^{K} \max\left(0, 1 - y^{(i)}(w^T x^{(i)} + b)\right)$$

(1)

① if $y^{(i)}(w^T x^{(i)} + b) \geq 1$: $\nabla_w \text{hinge}_{y^{(i)}}(x^{(i)}) = 0 \in \mathbb{R}^d$

② if $y^{(i)}(w^T x^{(i)} + b) < 1$: $\nabla_w \text{hinge}_{y^{(i)}}(x^{(i)}) = -y^{(i)} x^{(i)} \in \mathbb{R}^d$

Hadamard product

$$\Rightarrow \nabla_w \text{hinge}_{y^{(i)}}(x^{(i)}) = \mathbb{1}_{\{y^{(i)}(w^T x^{(i)} + b) < 1\}} \odot (-y^{(i)} x^{(i)})$$

$$\therefore \nabla_w L = \frac{1}{K}\sum_{i=1}^{K} \mathbb{1}\{y^{(i)}(w^T x^{(i)} + b) < 1\} \odot (-y^{(i)} x^{(i)})$$

(2)

① if $y^{(i)}(w^T x^{(i)} + b) \geq 1$: $\nabla_b \text{hinge}_{y^{(i)}}(x^{(i)}) = 0$

② if $y^{(i)}(w^T x^{(i)} + b) < 1$: $\nabla_b \text{hinge}_{y^{(i)}}(x^{(i)}) = -y^{(i)}$

$$\therefore \nabla_b L = \frac{1}{K}\sum_{i=1}^{K} \mathbb{1}\{y^{(i)}(w^T x^{(i)} + b) < 1\} \odot (-y^{(i)})$$